Capture-based VR Content Creation

캡처 기반 VR 모델링

가상현실 강의

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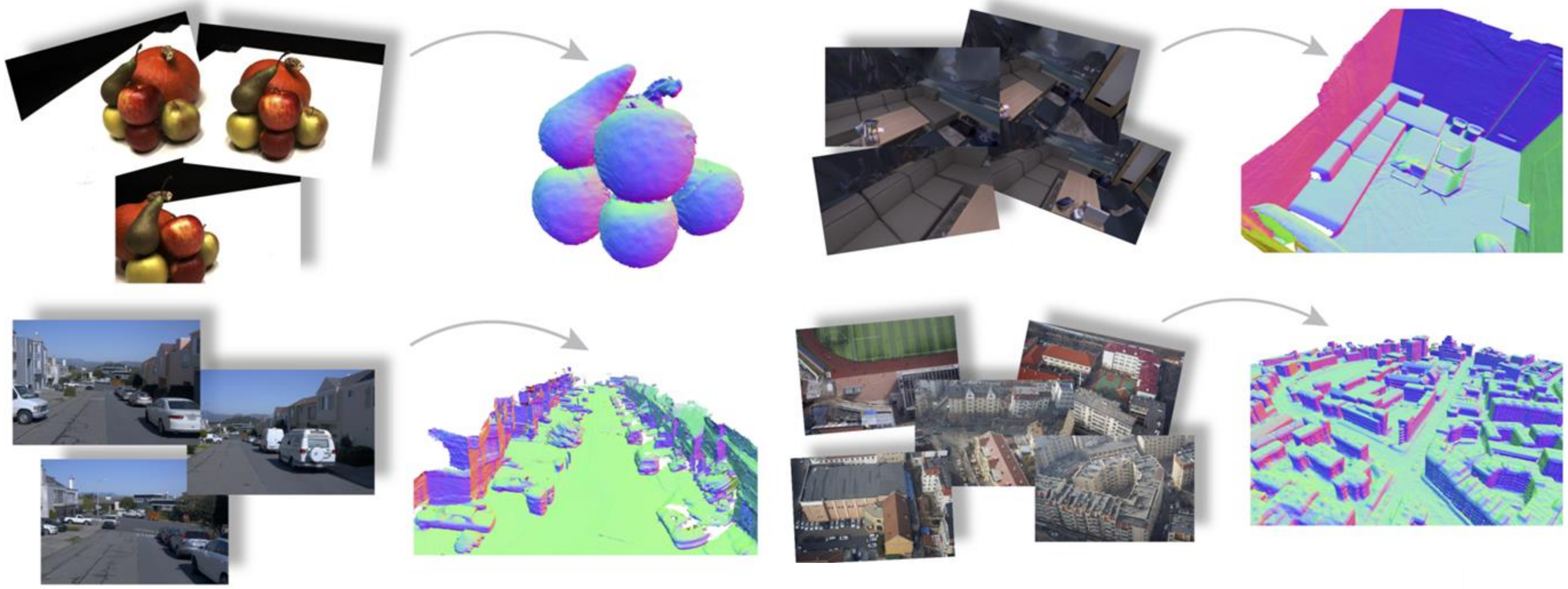
01.

Multi-view Stereo (MVS)

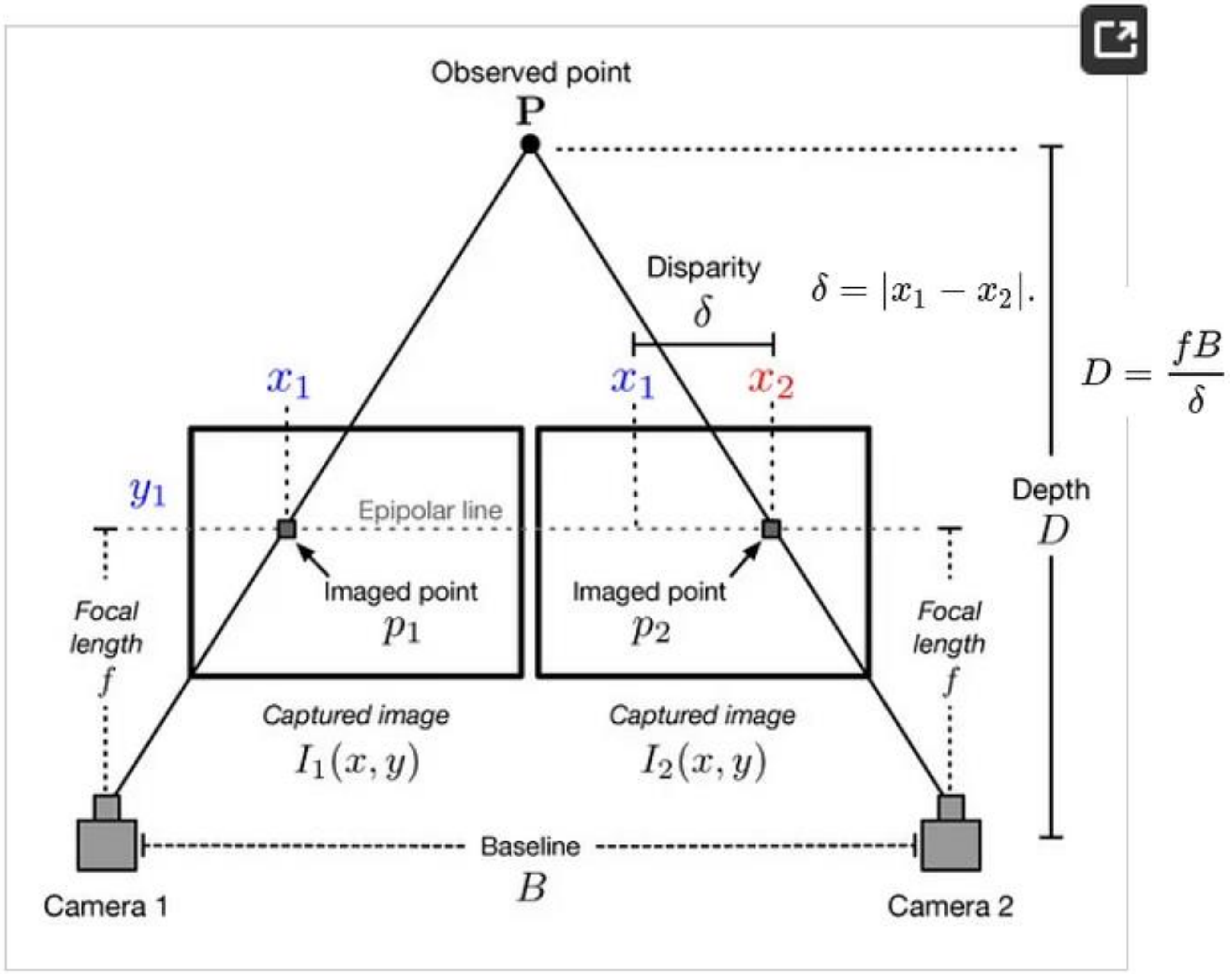
1 여러 시점에서 촬영한 이미지 입력



01.

Multi-view Stereo (MVS)

2 Stereo Matching을 통해 깊이 계산



Relationship between Depth Estimation and Disparity in case of Stereo Image setup

01.

Multi-view Stereo (MVS)

3 Dense depth map → Dense point cloud 생성

01.

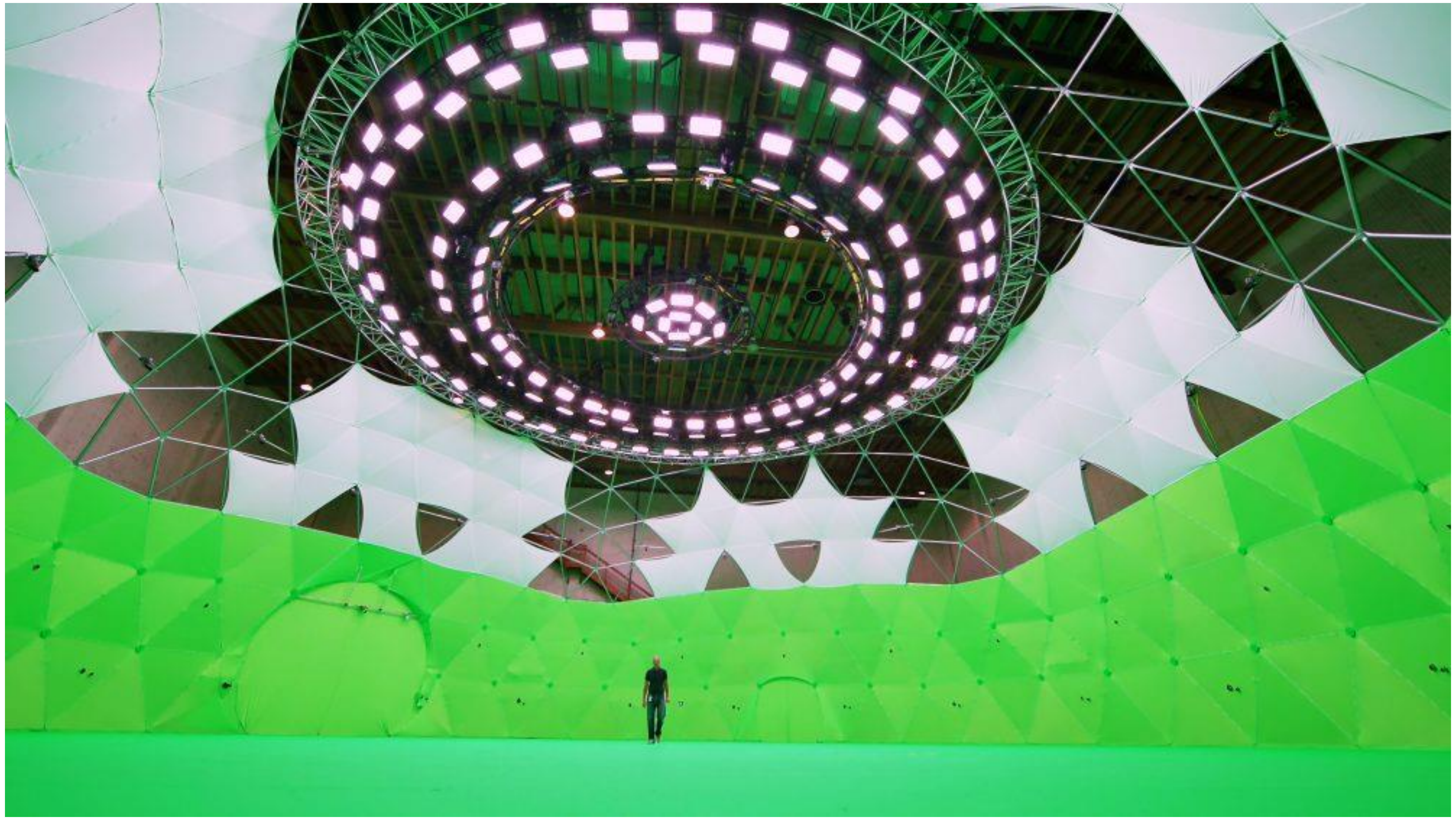
Multi-view Stereo (MVS)

4 포토그래메트리의 핵심 기술

02.

Volumetric Video

1 여러 대의 카메라가 동기화되어 촬영



Intel Studios' Volumetric Capture Space

02.

Volumetric Video

2 디지털 휴먼·공연·스포츠 실시간 재현 가능

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02.

Volumetric Video

3 Meta·Microsoft·Intel 등이 볼류메트릭 스튜디오 운영

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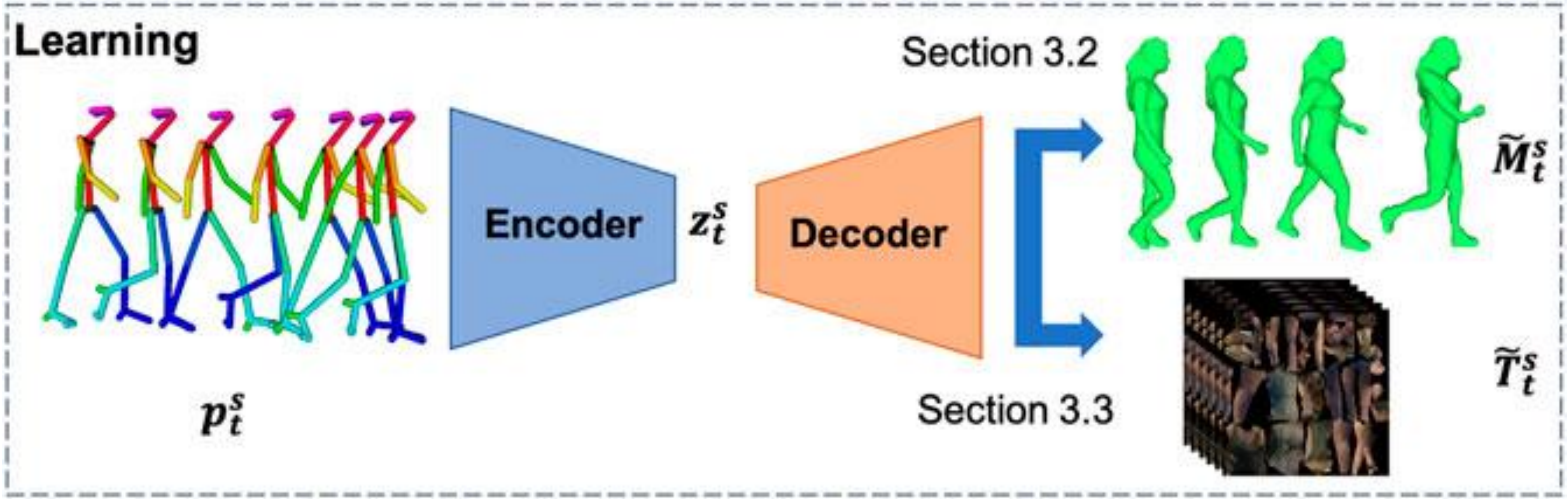
02.

Volumetric Video to Mesh

4 시간에 따라 변하는 Mesh sequence 생성



03. Mesh Fusion (TSDF / Poisson)

TSDF (Truncated Signed Distance Function)

- KinectFusion에서 사용된 핵심 기술
  - 깊이 센서 데이터를 지속적으로 누적
  - 공간을 3D voxel grid로 표현하면서 표면 위치를 거리 함수로 추정
- 실시간 매끄러운 표면 생성
  - 센서 노이즈를 자연스럽게 보정
  - 움직이면서 스캔해도 실시간으로 부드러운 Mesh 생성
  - VR/AR용 실시간 SLAM·환경 매핑에 최적화

포인트클라우드를 Mesh로 변환하는 기술

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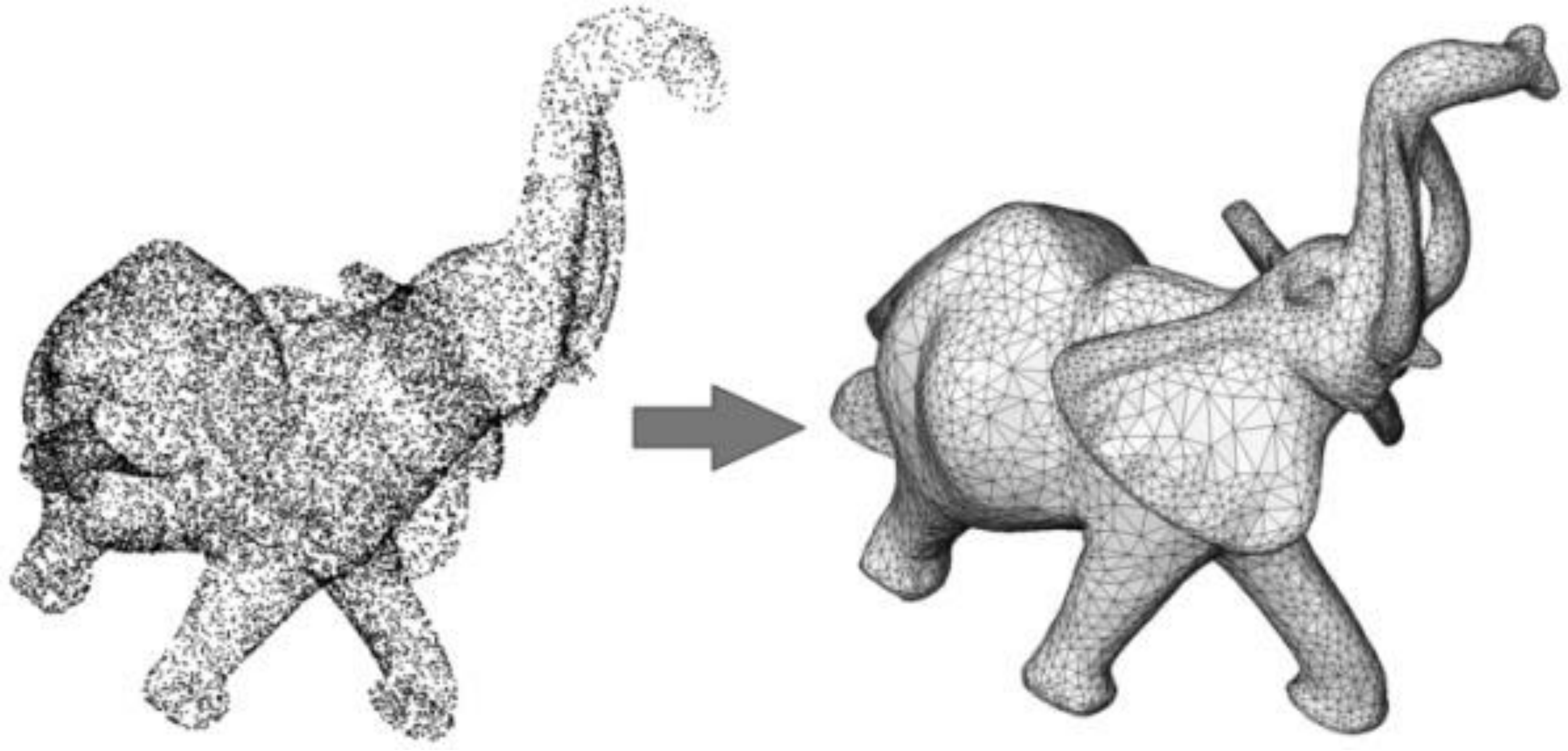
03. Mesh Fusion (TSDF / Poisson)

Poisson Reconstruction

- 정밀한 Surface 재생성 알고리즘
  - 포인트 클라우드 + 노멀을 이용해 물체 표면을 전역적으로 추론
  - 작은 구멍도 자동 보완 → 매우 매끄럽고 연속적인 Mesh 출력

- 모델링 퀄리티 최고 수준
  - 포토그래메트리·스캔 데이터 후처리에 널리 사용
  - CAD·CG 분야에서도 고품질 Mesh 생성의 표준 방식



포인트클라우드를 Mesh로 변환하는 기술

03. 캡처 기반 데이터 너무 많음 → 어떻게 표현?

캡처 데이터의 문제

- 수백만 포인트
- Volumetric Video는 TB 단위
- 그대로 VR 불가

표현 전략

- Downsampling / Voxel grid
- TSDF
- Gaussian Splatting
- Compression

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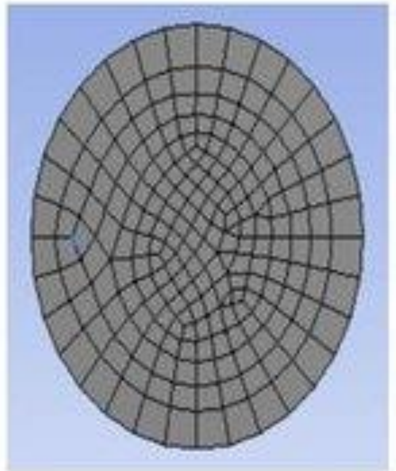
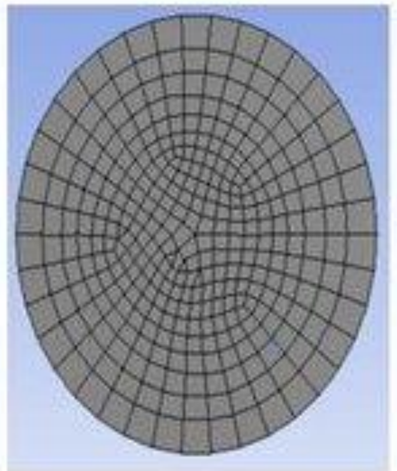
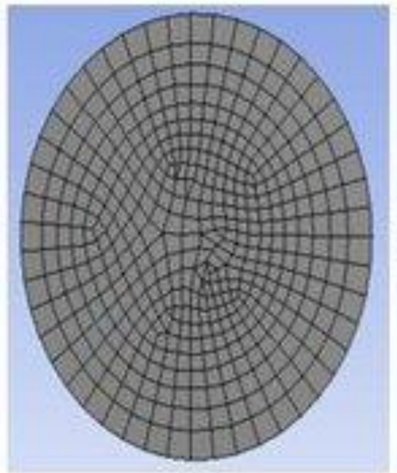
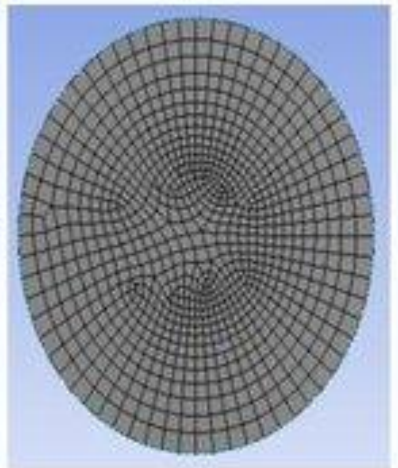
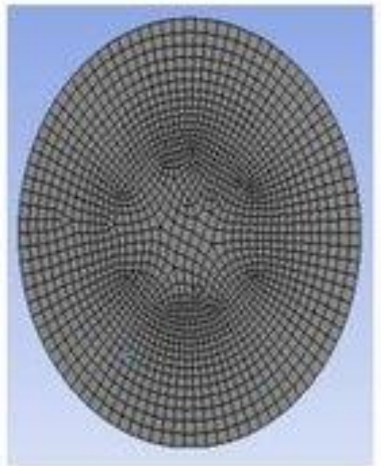
04.

Time-varying

Mesh Compression

1 매우 무거운 데이터 구조

- 한 프레임이 하나의 완전한 3D Mesh
- 프레임 수가 많아지면 용량이 폭발적으로 증가 (GB~TB 단위)

Mesh	1	2	3	4	5
Element Size (m)	0.03	0.025	0.02	0.015	0.01
Number of Elements	48428	75651	115739	272062	454805
Mesh Visualization on Outlet					

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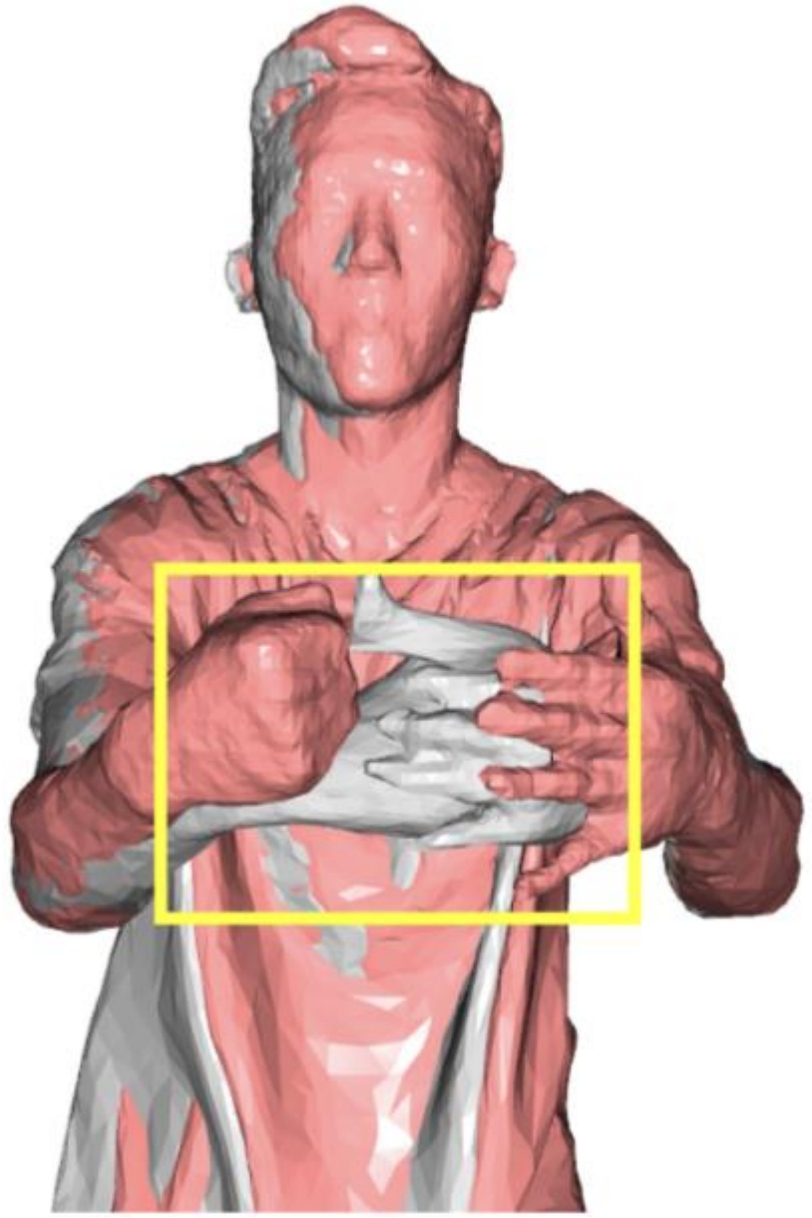
Mesh Compression

2 시간에 따라 변하는 Mesh의 구조

- 정점(Vertex) 위치가 매 프레임 변경
  - 애니메이션이 아니라 진짜 'Mesh 자체'가 변형됨
  - 복잡한 사람/옷/머리카락 움직임까지 모두 포함



04.

Time-varying

Mesh Compression

3

효율적 전송을 위한 압축 방식

- Frame-to-frame Delta 압축
- 매 프레임 전체를 저장하지 않고
  - "이전 프레임에서 얼마나 변했는지"만 저장
- Draco 압축 활용
  - Google의 기하/속성 압축
  - WebXR·WebGL에서도 표준적으로 사용됨

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04.

Time-varying

Mesh Compression

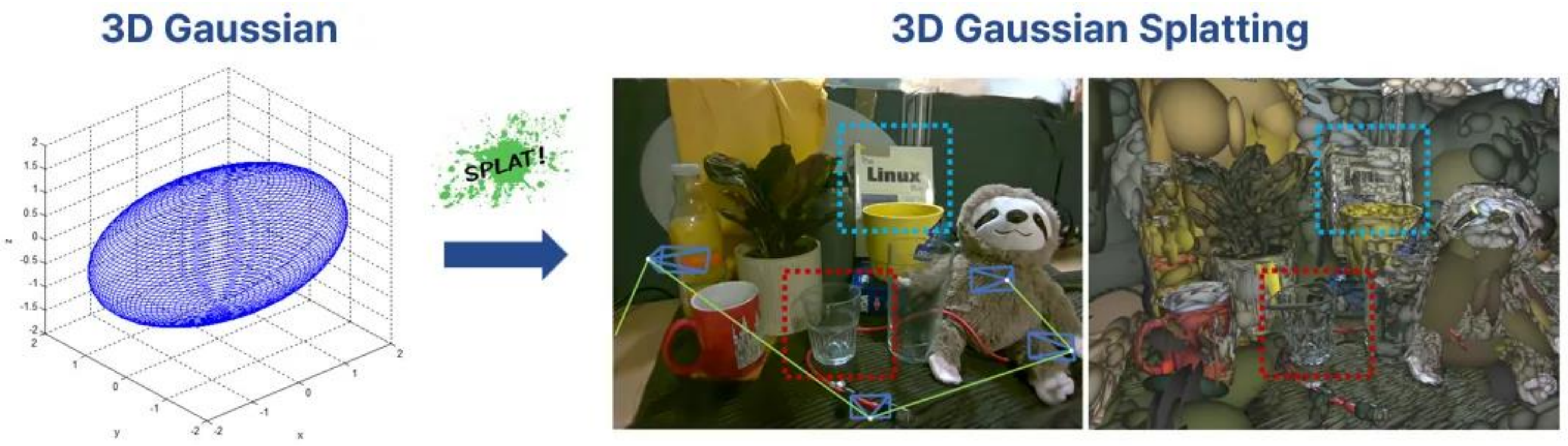
4 VR 스트리밍에서 필수 기술

- 실시간 전송을 위해 대용량 Mesh를
  - → 빠르게 압축
  - → 빠르게 디코딩
  - → GPU에 즉시 업로드
- Meta, Google, Microsoft 등이 모두 Volumetric Video 스트리밍 기술 개발 중

31.

Gaussian Splatting – based Representation

1. 입력: 포인트 클라우드 기반
2. 각 포인트를 3D Gaussian(Ellipsoid)로 최적화
3. Mesh 없이도 고품질 3D 장면 표현 가능
4. MVS/Depth 기반 캡처와 궁합이 매우 좋음



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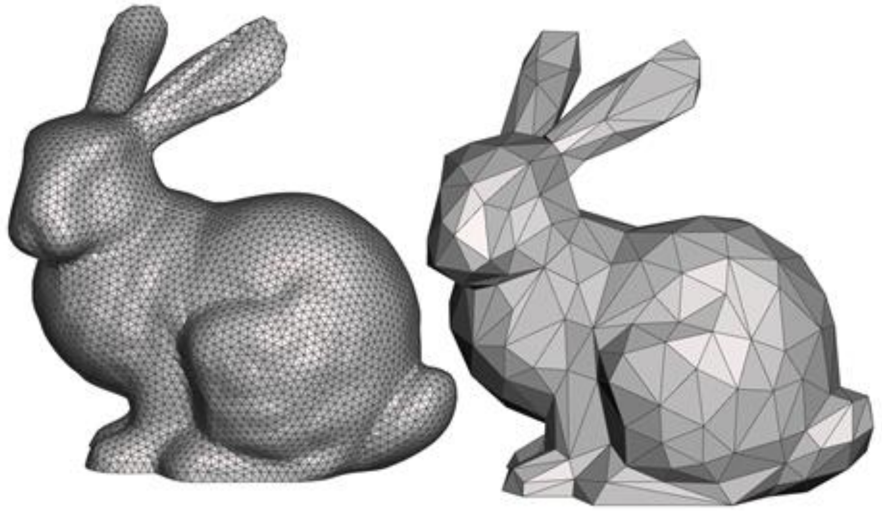
1 영상 기반

- 장점: 현실감 최고
- 단점: 3DoF 제한
- 대표 기술: Panorama, NeRF

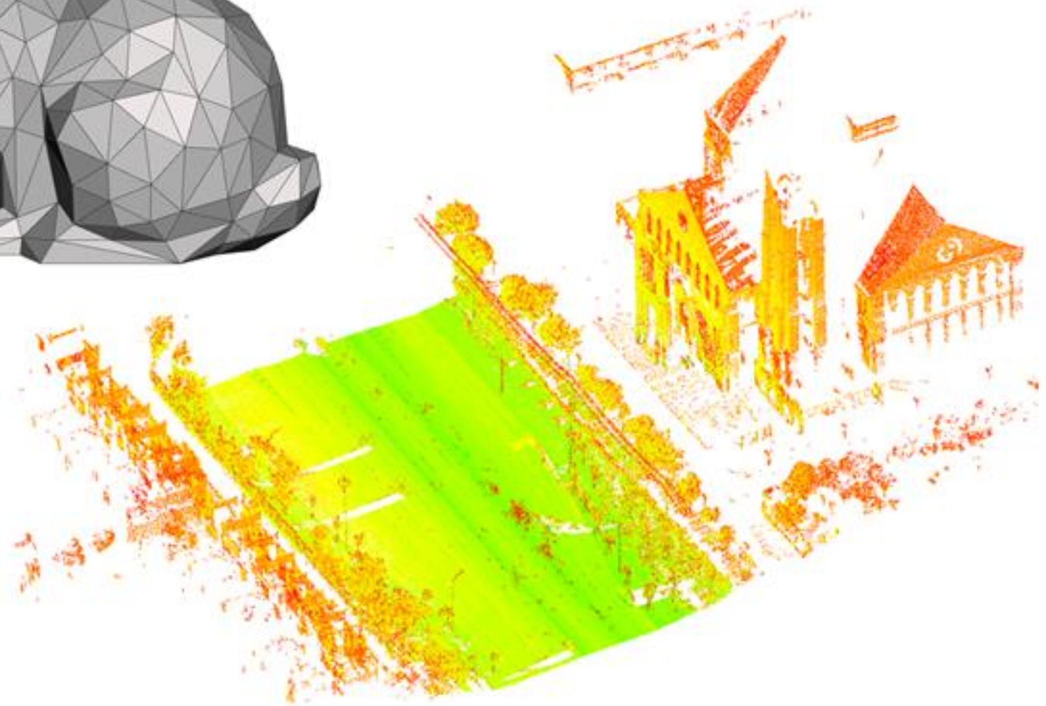
2 모델링 기반

- 장점: 상호작용/6DoF
- 단점: 제작비와 인력 소요
- 대표 기술: Mesh/LOD



3 캡처 기반

- 장점: 실제 공간/사람 재현
- 단점: 스튜디오와 장비 비용
- 대표 기술: MVS, Volumetric



32. 캡처 기반 vs 영상 기반 vs 모델링 기반 비교

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요약

1. Capture-based VR은 실제 세계를 3D로 가져오는 기술
2. Point Cloud → Mesh → Neural 방식으로 발전
3. Volumetric video는 미래 디지털 휴먼 구현의 핵심
4. Gaussian Splatting이 새로운 실시간 3D 캡처 기반 기술로 정착 중

이미지 출처
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감사합니다

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